

Amedeo Chiefa

Ph.D. Candidate, Theoretical Particle Physics

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Research Interests

I am a Ph.D. candidate (2023–2027) working at the interface of QCD phenomenology and machine learning in the determination of parton distribution functions from experimental data. My research develops a quantitative understanding of how deep neural networks learn parton distributions: I use neural tangent kernel theory to characterise the training dynamics of PDF fits and to establish whether the uncertainties such fits report are faithful. I am an active member developer in the NNPDF collaborations, with six publications since 2024 in *PLB*, *JHEP* and *EPJC*.

Publications

Author lists in this field follow the alphabetical convention. Publications in which I led the analysis are marked †.

First next-to-next-to-leading-order extraction of fragmentation functions for Lambda hyperons

V. Bertone, A. Bongallino, **A. Chiefa**, M. G. Echevarria, G. Schnell

Submitted to JHEP (under review) | e-Print: arXiv:2605.05314

Quantitative Understanding of PDF Fits and their Uncertainties †

A. Chiefa, L. Del Debbio, R. Kenway

Published in: *Eur.Phys.J.C* 86 (2026) 6, 722 | e-Print: arXiv:2512.24116

Parton distributions with higher twist and jet power corrections †

R. D. Ball, **A. Chiefa**, R. Stegeman

Published in: *Eur.Phys.J.C* 86 (2026) 3, 281 | e-Print: arXiv:2511.14387

Parton distributions confront LHC Run II data: a quantitative appraisal

A. Chiefa, M. N. Costantini, J. Cruz-Martinez, E. R. Nocera, T. R. Rabemananjara, J. Rojo, T. Sharma, R. Stegeman, M. Ubiali

Published in: *JHEP* 07 (2025) 067 | e-Print: arXiv:2501.10359

Helicity-dependent parton distribution functions at NNLO accuracy from inclusive and semi-inclusive deep-inelastic scattering data †

V. Bertone, **A. Chiefa**, E. R. Nocera

Published in: *Phys.Lett.B* 865 (2025) 139497 | e-Print: arXiv:2404.04712

Status and Developments in Polarised Parton Distribution Functions †

A. Chiefa

PoS DIS2024 (2025) 200

Education

Ph.D. in Theoretical Physics

11/2023 – expected 10/2027

The University of Edinburgh (UK) — Supervisor: Prof. Luigi Del Debbio

Funded by the Higgs Fellowship

M.Sc. in Theoretical Physics (110/110 cum laude)

24/10/2023

University of Turin — Supervisor: Prof. Emanuele R. Nocera

Thesis: Towards Polarised Parton Distribution Functions at next-to-next-to-leading order

B.Sc. in Physics (109/110)

20/07/2021

University of Turin — Supervisor: Prof. Marco Regis

Thesis: Self-interacting dark matter: cross-section calculation

Conference Contributions & Invited Talks

Parton Distributions from NTK analysis

10/2025

Invited at Digital Twins for Nuclear and Particle Physics 2025

Parton Distributions from NTK analysis

09/2025

COMETA Workshop on Uncertainty Quantification for Machine Learning

Towards helicity-dependent parton distribution functions at NNLO accuracy

04/2024

31st International Workshop on Deep Inelastic Scattering (DIS2024)

Regular internal presentations at NNPDF collaboration meetings (2023–present).

Research Experience and Collaborations

NNPDF Collaboration — Member and Developer

since 11/2023

Contribute to the maintenance and development of the NNPDF fitting codebase, a production Python framework

used collaboration-wide to determine the parton distribution functions of the proton.

MAP Collaboration — Developer since 03/2023
Co-developed the codebase Denali that extended the MAP analysis framework to polarised parton distribution functions; currently extending it to higher-order theoretical accuracy.

CEA Paris-Saclay, IRFU (France) — Erasmus+ Traineeship 03/2023 – 05/2023
Supervisor: Dr Valerio Bertone. Determination of the polarised parton distribution functions of the proton at NNLO within the MAP framework. Competitively funded by the European Commission.

Awards, Funding and Training

Iliad Intensive, LISA London 06/2026 – 07/2026
Month-long selective taught course in graduate-level applied mathematics for AI alignment research.

AI for Science European Summer School 07/2026
Applications of AI in scientific research; sponsored by Google DeepMind and the Hellenic Institute of Advanced Studies.

Teaching

School of Physics and Astronomy, University of Edinburgh (UK): **Teaching Assistant** in Statistical Physics (4th year, 2025/2026); **Teaching Assistant** in Principles of Quantum Mechanics (3rd year, 2024/2025 and 2025/2026, also teaching duties); **Teaching Assistant** in Computer Simulation (2nd year, 2023/2024).

Technical Skills

Programming: Python (primary), C++, C, Fortran, Bash
Machine learning & numerics: PyTorch, TensorFlow, JAX, NumPy, SciPy
Methods: neural tangent kernel analysis, uncertainty quantification, Monte Carlo methods, regression and model fitting on correlated experimental data, closure testing, perturbative QCD and collider physics
Software engineering: Git, GitHub Actions (CI/CD), code review in a distributed collaboration, pytest, conda, HPC/SLURM
Other: Mathematica, L^AT_EX

Languages and Referees

Languages: Italian (native); English (full professional proficiency); French (basic).
Referees: Prof. Luigi Del Debbio (University of Edinburgh — Ph.D. supervisor); Prof. Richard Kenway (University of Edinburgh — co-author, *EPJC* 86, 722); Prof. Emanuele R. Nocera (University of Turin — M.Sc. supervisor); Dr Valerio Bertone (CEA Paris-Saclay, IRFU — Erasmus+ traineeship supervisor). Contact details available on request.